Matlab Code

```
clc
clear
close all

% Load data
load 'HBP_Input_Stance.mat'
N = length(COP);


%define mass and moments of inertia for segments
%foot
IGf=0.0037;
mf=0.86;
%shank
ms=2.73;
IGs=0.0237;
%thigh
mt=8.92;
IGt=0.1903;

% Initialize calculated arrays
%ankle
Rax = zeros(N, 1); % Creates an array of zeros with N rows and 1 column
Raz = zeros(N, 1);
Ma = zeros(N, 1);
%knee
Rkx = zeros(N, 1);
Rkz = zeros(N, 1);
Mk = zeros(N, 1);
%hip
Rhx = zeros(N, 1);
Rhz = zeros(N, 1);
Mh = zeros(N, 1);

% Time array
time = zeros(N,1);
for i = 1:N
    time(i,1) = (time(i,1) + i)/100;
end

% Calculate the ankle rxn forces and moments
for i = 1:N

    aGxf = foot_COM_la(i,1); % Accesses the i-th row and 2nd column
    aGzf = foot_COM_la(i,2);
    alphaf = foot_aa(i); % Only need to specify 1D because the array is 1D
    GRFx = GRF(i,1);
    GRFz = GRF(i,2);
    xa = RANKLE(i,1);
    za = RANKLE(i,1);
    xGf = foot_COM(i,1); % x pos of COM
    zGf = foot_COM(i,2); % z pos of COM
    COPx = COP(i,1);
    COPz = COP(i,2);
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    Raz(i)=(mf*aGzf)+(mf*9.81)-GRFz;
    Ma(i)=(IGf*alphaf)+(Rax(i)*(za-zGf))+(Raz(i)*(xGf-xa))-(GRFz*(COPx-xGf))-(GRFx*(zGf));

% Calculate the hip rxn forces and moments
    aGxs = shank_COM_la(i,1);
    aGzs = shank_COM_la(i,2);
    alphas = shank_aa(i);
    xk = RKNEE(i,1);
    zk = RKNEE(i,2);
    xGs = shank_COM(i,1);
    zGs = shank_COM(i,2);

    Rkx(i)=(ms*aGxs)+Rax(i);
    Rkz(i)=(ms*aGzs)+Raz(i)+(ms*9.81);
    Mk(i)=Ma(i)+(IGs*alphas)+(Rkx(i)*(zk-zGs))+(Rkz(i)*(xGs-xk))+(Rax(i)*(zGs-za))+(Raz(i)*(xa-xGs));

% Calculate the thigh rxn forces and moments
    aGxt = thigh_COM_la(i,1);
    aGzt = thigh_COM_la(i,2);
    alphat = thigh_aa(i);
    xh = RTROCH(i,1);
    zh = RTROCH(i,2);
    xGt = thigh_COM(i,1);
    zGt = thigh_COM(i,2);

    Rhx(i)=(mt*aGxt)+Rkx(i);
    Rhz(i)=(mt*aGzt)+Rkz(i)+(mt*9.81);
    Mh(i)=Mk(i)+(IGt*alphat)+(Rhx(i)*(zh-zGt))+(Rhz(i)*(xGt-xh))+(Rkx(i)*(zGt-zk))+(Rkz(i)*(xk-xGt));

end


% graph the ankle rxn forces and moments
figure(1)
plot(time, Rax,'color','#FF00FF');
title('Reaction Forces in X-Direction Over Time','fontsize',14)
hold on
plot(time, Rkx,'color','#0DFF00');
plot(time, Rhx,'color','#0000FF');
legend('Ankle','Knee','Hip')
ylabel('Reaction Forces (N)','fontsize',14)
xlabel('Time (s)','fontsize',14)
grid on
hold off

figure(2)
plot(time, Raz,'color','#FF00FF');
title('Reaction Forces in Z-Direction Over Time','fontsize',14)
hold on
plot(time, Rkz,'color','#0DFF00');
plot(time, Rhz,'color','#0000FF');
legend('Ankle','Knee','Hip')
xlabel('Time (s)','fontsize',14)
ylabel('Reaction Forces (N)','fontsize',14)
grid on
hold off
```

```
subplot(3, 1, 1);
plot(time, Ma,'color','#FF00FF');
legend('Ankle')
%xlim([0,.8]);
%ylim([-100,0]);
grid on
title('Joint Moments Over Time','fontsize',14)
subplot(3, 1, 2);
plot(time, Mk,'color','#0DFF00');
legend('Knee')
%xlim([0,.8]);
%ylim([-20,20]);
grid on
ylabel('Joint Torque (N*m)','fontsize',14)
subplot(3,1, 3);
plot(time, Mh,'color','#0000FF');
legend('Hip')
%xlim([0,.8]);
%ylim([-20,40]);
grid on
xlabel('Time (s)','fontsize',14)
```